EXPERIMENT 11: ROUTING AT NETWORK LAYER

Aim: To simulate Static Routing and RIP (Routing Information Protocol) using Cisco Packet Tracer.

A. Static Routing

Theory: Static routes are manually added to a router's routing table. A route with a lower administrative distance (AD) is preferred. A higher-AD route can act as a backup if the main route fails.

Procedure

1. Create the three-router topology shown in the lab manual, with networks 10.0.0.0/8, 20.0.0.0/8, 30.0.0.0/8, 40.0.0.0/8 and 50.0.0.0/8 connected across the routers.

2. Assign IP addresses to router interfaces and PCs, and enable interfaces using `no shutdown`.

3. Add static routes for networks not directly connected to each router. Use AD 10 for the main route and AD 20 for the backup route where required.

Router0 — static route configuration

```
enable
configure terminal
ip route 30.0.0.0 255.0.0.0 20.0.0.2 10
ip route 30.0.0.0 255.0.0.0 40.0.0.2 20
ip route 30.0.0.100 255.255.255.255 40.0.0.2 10
ip route 30.0.0.100 255.255.255.255 20.0.0.2 20
ip route 50.0.0.0 255.0.0.0 40.0.0.2 10
ip route 50.0.0.0 255.0.0.0 20.0.0.2 20
end
show ip route static
```

Sample output

```
S 30.0.0.0/8 [10/0] via 20.0.0.2
S 30.0.0.100/32 [10/0] via 40.0.0.2
S 50.0.0.0/8 [10/0] via 40.0.0.2
```

The routing table contains the preferred static routes. Backup routes remain available and are used if the preferred path fails.

Verification: Ping between end hosts and use `tracert` from a PC to observe the path. Disconnect the main link and test again to verify backup routing.

B. RIP (Routing Information Protocol)

Theory: RIP is a dynamic distance-vector routing protocol. Routers exchange information about directly connected networks and select routes using hop count.

IP addressing table from the lab manual

Device	Interface	IP address / prefix
PC0	FastEthernet	10.0.0.2/8
Router0	Fa0/1	10.0.0.1/8
Router0	S0/0/0	192.168.1.249/30
Router1	S0/0/0	192.168.1.250/30
Router1	S0/0/1	192.168.1.246/30
Router2	S0/0/0	192.168.1.245/30
Router2	S0/0/1	192.168.1.253/30
Router0	S0/0/1	192.168.1.254/30
Router2	Fa0/1	20.0.0.1/8
PC1	FastEthernet	20.0.0.2/8

Procedure

1. Assign the listed IP addresses to PCs and router interfaces. Enable each interface with `no shutdown`. Set a clock rate on serial interfaces only at the DCE end.

2. Configure RIP on each router and advertise its directly connected networks.

Router0 — RIP configuration

```
enable
configure terminal
router rip
network 10.0.0.0
network 192.168.1.252
network 192.168.1.248
end
```

Router1 — RIP configuration

```
enable
configure terminal
router rip
network 192.168.1.244
network 192.168.1.248
end
```

Router2 — RIP configuration

```
enable
configure terminal
router rip
network 20.0.0.0
network 192.168.1.252
network 192.168.1.244
end
```

Testing connectivity

```
PC0> ping 20.0.0.2
Reply from 20.0.0.2: bytes=32 time<1ms TTL=126
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Successful replies indicate that RIP has learned a route between the two networks. Use `tracert 20.0.0.2` to view the selected path. Disconnect the direct Router0–Router2 serial link and repeat the test; RIP should reroute traffic through

Router1.

Observations

1. Static routing requires routes to be entered manually; RIP exchanges route information automatically.
2. Static routes can be configured with a backup administrative distance. RIP selects a path based on hop count and can adapt when a route fails.

Result: Static routing and RIP were configured in Cisco Packet Tracer. Connectivity and route selection were verified using ping and traceroute commands.